LALAN PRASAD RAI

M.Sc. in Electrical Engineering (Power Electronics) | NIT Jamshedpur

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EDUCATION

M.Sc. in Electrical Engineering (Specialization: Power Electronics) — National Institute of Technology (NIT), Jamshedpur, Jharkhand

- Core areas: Thyristor Circuits I & II, Control Theory, Instrumentation, Microprocessor Systems, Advanced Mathematics

Bachelor of Engineering — AMIE (Electronics & Communication Engineering) — Institution of Engineers (India), Jamshedpur

- Section A: Mathematics, Applied Mechanics & Strength of Material, Material Science, Electrical & Mechanical Engineering
- Section B: Pulse & Digital Circuits, Circuit Theory, Electronic Devices & Circuits, Communication Systems, Industrial Electronics, Computer Engineering, Electronic Measurements & Instruments

Bachelor of Legislative Law (LL.B.) — Netaji Subhas University, Jamshedpur (Session: 2020–23; Exam: July 2023)

- Result: **First Class**

Diploma in Electronics & Communication Engineering — New Govt. Polytechnic, Patna | State Board of Technical Education, Bihar

Matriculation — S.R.C. High School, Patepur, Vaishali, Bihar | Bihar School Examination Board, Patna

PROFESSIONAL MEMBERSHIPS & LICENSURE

Associate Member — Institution of Engineers (India) [AMIE]

Government License — Licensed by the Jharkhand Government for Electrical AC Installation up to 33 KV

AREAS OF TEACHING COMPETENCE

- Power Electronics and Industrial Drive Systems
- Programmable Logic Controllers and Industrial Automation
- Circuit Theory and Electronic Devices
- Instrumentation, Measurement, and Control Systems
- Communication Engineering
- Microprocessor and Microcontroller Systems
- Engineering Law and Regulatory Frameworks (leveraging LL.B., First Class)

TEACHING EXPERIENCE

Tata Steel — SNTI (School of New Technology & Innovation), Jamshedpur

Part-time Faculty, Diploma Courses

1998 – 2007 (9 Years)

- Designed and delivered rigorous instruction in Circuit Theory, Instrumentation, Communication Engineering, and Computer Engineering to diploma-level engineering students, consistently integrating real-world industrial applications into course content over nine consecutive years (1998–2007).
- Mentored working engineering professionals in applied problem-solving, translating the complexity of live industrial systems — including PLCs, drives, and process control — into structured, pedagogically coherent academic material.

Bihar State Electronics Development Corporation Ltd., Danapur (Special Projects), Patna

Technical Trainer

May 1985 – June 1986

- Designed and facilitated structured instruction on Television theory and analog electronic circuits for trainee cohorts, emphasizing conceptual clarity and applied understanding.

- Led hands-on PCB-level laboratory sessions on television circuit systems, reinforcing theoretical foundations through guided, applied demonstrations and diagnostic exercises.

INDUSTRY EXPERIENCE

Tata Steel, Jamshedpur — Electrical Maintenance Division

Electrical & Electronics Maintenance Engineer

July 1986 – November 2025 (Retired)

- Commissioned and maintained Siemens S7 PLC systems for Gas Cleaning Plant operations, ensuring sustained process reliability across high-criticality environmental control subsystems.
- Architected full-cycle Siemens 135U PLC software — encompassing hydraulics, pneumatics, interlocking logic, and annunciation — for Wire and Rod Mill furnace operations, eliminating manual intervention at critical control points.
- Commissioned a four-quadrant, fully controlled DC drive (AVTRON, 3,500 HP) for the main mill motor at Bar Mill, enabling precise speed and torque regulation for heavy industrial rolling operations.
- Commissioned Schneider Twido Soft PLC and Control Technique UNI Drive SP Digital Drive for elevator control systems, improving motion accuracy and operational safety.
- Commissioned microprocessor-based soft-starter system for pump motor control at Gas Cleaning Plant (GCP), reducing electrical stress at startup and extending motor service life.
- Commissioned and maintained microprocessor-based weighing systems across platform (40–60 T), rail (80 T), and in-motion (100 T) configurations, ensuring measurement accuracy critical to production quality control.
- Diagnosed and resolved complex faults across PLCs, AC/DC drives, UPS systems, CNC machines, high-power induction heaters, and DAS/SCADA platforms, minimizing unplanned downtime across critical plant systems.
- Orchestrated inter-departmental maintenance scheduling during equipment failures, coordinating rapid cross-functional response to restore production continuity under time-critical conditions.

- Administered spare parts inventory and strategic procurement planning to sustain operational readiness across a large, multi-system electrical maintenance division.
- Developed and maintained comprehensive maintenance check sheets, calibration records, and compliance documentation in strict adherence to ISO 9000 and ISO 14000 standards.
- Spearheaded QC circles, Value Engineering (VE) initiatives, and Quality Improvement Plans (QIP), contributing to measurable gains in system efficiency and cost reduction.

TECHNICAL EXPERTISE

PLC Systems	Siemens S7 / 135U — full-cycle integration including software development, commissioning, and maintenance; also Suco's, CP-80, Texas Instruments, Modicon, GE Fanuc, Honeywell, Allen Bradley (AB)
DC Drives (Analog)	Nelco, Siemens, BBC — speed and torque regulation for heavy industrial motors
DC Drives (Digital)	Control Technique, Avtron (up to 3,500 HP four-quadrant), Eurotherm
AC Drives	Control Technique UNI Drive SP (Digital); SAMI-B Stromburg (Analog) — variable-frequency drive commissioning and fault diagnosis
UPS Systems	Nea Lindburge (2×5 KVA), Semidan (2×10 KVA), Tata Liebert (2×10 KVA), Emerson (2×10 KVA)
Induction Heaters	AEG Elotherm, Germany (1.2 KV, 1.9 KHz, 2.6 MW); IPE Cheston, USA (1.2 KV, 180 Hz, 1 MW) — high-power industrial heating systems

Weighing Systems	Philips, Sartorius, Avery, Carl Schenk, Senlogic — Platform (40–60 T), Rail (80 T), In-Motion (100 T); calibration and metrology compliance
Process Control	Oxygen Plants (1×500 TPD, 2×275 TPD), Lime Calcining Plant (5×300 TPD) — large-scale continuous process automation
CNC Systems	AEG Telefunken 4-Axis Control System — commissioning, programming, and maintenance
DAS & SCADA	Intellution FIX, Coros 2000, Scan 3000 (Honeywell), Windows NT — real-time data acquisition and supervisory control

LANGUAGES & INTERESTS

Languages: Hindi (Native), English (Proficient)

Date of Birth: 04 October 1965

Interests: Trekking, Carpentry,